Analysis of Factors Associated with Poor Response to Nutritional Treatment in Severely Malnourished Children Aged 6–59 Months With Edema and With Complications Treated (Phase 1).

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1. Methodology

Descriptive analyses were conducted to characterize the cohort and the non-response to the nutritional treatment. Continuous variables were summarized using medians and interquartile ranges (25th - 75th percentile) due to their non-normal distribution, whereas categorical variables were summarized as counts and percentages (%).

Differences between groups (success and failure) were assessed using the Wilcoxon rank-sum test or Kruskal–Wallis test for continuous variables and Pearson's chi-square or Fisher's exact test for categorical variables, as appropriate.

Univariable logistic regression analyses were performed to estimate crude associations between independent variables and the dependent variable. Variables considered clinically relevant or statistically associated with the non-response to the nutritional treatment at $\underline{p} < 0.20$ were included in a multivariable logistic regression.

Prior to model building, pairwise correlations among independent variables were examined to identify highly correlated predictors and assess potential multicollinearity. Pearson correlation coefficients were computed for continuous variables (or Spearman's rank correlations for non-normally distributed variables). Variables exhibiting strong correlations ($|r| \geq 0.70$) were carefully evaluated to avoid redundancy. Variance Inflation Factors (VIFs) were subsequently calculated in the multivariable model, with VIF values > 5 indicating potential multicolliniarity.

Model selection was guided by likelihood ratio tests and the principle of parsimony. A stepwise selection procedure was used to identify the most parsimonious model (based on the Akaike Information Criterion (AIC)) by retaining variables that improved model fit while excluding predictors that contributed little additional information.

Adjusted odds ratios (aORs) with 95% confidence intervals (CIs) were reported. Statistical significance in the final model was defined as a two-sided $\underline{p} < 0.05$. A variable was significantly associated with the risk of nutritional treatment failure if: i) the odds ratio (OR) was > 1 with a lower limit of the confidence interval (CI) > 1; or ii) OR<1 with upper CI limit <1; and p-value was <0.05.

All analyses were conducted using R (version 4.3.0).

2. Study population

Insert comments here.

3. Descriptive Characteristics

Insert comments here.

4. Bivariate Analysis

Table 1: Association between explanatory variables and outcome at phase 1.

	Non-response to treatment in Phase				
Characteristic	**FALSE** N = 147[1]	**TRUE** N = 117[1]	**Overall** N = 264[1]	**p-value**[23]	**q-value**[4]
Age (Month)				0.720	0.790
Median (Q1 ; Q3)	24 (18.0 ; 32)	27 (19.0 ; 34)	26 (18.5 ; 32)		
Min ; Max	7 ; 59.0	8 ; 59.0	7 ; 59.0		
Age group				0.517	0.701
[25;59]	72 (49.0%)	62 (53.0%)	134 (50.8%)		
[6;25[	75 (51.0%)	55 (47.0%)	130 (49.2%)		
Gender				0.386	0.571
Male	82 (55.8%)	59 (50.4%)	141 (53.4%)		
Female	65 (44.2%)	58 (49.6%)	123 (46.6%)		
Locality				0.588	0.705
Urban	28 (19.4%)	19 (16.8%)	47 (18.3%)		
Rural	116 (80.6%)	94 (83.2%)	210 (81.7%)		
Missing values	3	4	7		
Vaccination status				0.514	0.701
Up to date	41 (75.9%)	39 (81.3%)	80 (78.4%)		
Out of date	13 (24.1%)	9 (18.8%)	22 (21.6%)		
Missing values	93	69	162		
Breastfeeding status	80 (89.9%)	58 (84.1%)	138 (87.3%)	0.274	0.446
Missing values	58	48	106		
Type admission				0.876	0.902
New admission	123 (87.9%)	100 (88.5%)	223 (88.1%)		
Readmission	17 (12.1%)	13 (11.5%)	30 (11.9%)		
Missing values	7	4	11		
Persistent edema in day5 (P1)	0 (0.0%)	59 (50.4%)	59 (22.3%)	**<0.001*** **	<0.001

Characteristic	Non-response to treatment in Phase				
	FALSE N = 147[1]	**TRUE** N = 117[1]	**Overall** N = 264[1]	**p-value**[2,3]	**q-value**[4]
WHZ error proportion (P1)				0.162	0.367
Median (Q1 ; Q3)	1 (1.0 ; 1)	1 (0.0 ; 1)	1 (1.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
Missing values	6	4	10		
Tube feeding duration (P1)				**0.002****	0.011
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 4.0	0 ; 10.0	0 ; 10.0		
Phase length (P1)				**<0.001*****	<0.001
Median (Q1 ; Q3)	3 (2.0 ; 5)	7 (5.0 ; 10)	4 (3.0 ; 7)		
Min ; Max	1 ; 18.0	1 ; 30.0	1 ; 30.0		
Nutritional treatment (P1)				**<0.001*****	<0.001
F75	107 (72.8%)	40 (34.2%)	147 (55.7%)		
F75F	24 (16.3%)	11 (9.4%)	35 (13.3%)		
F75-F75F/F75F-F75	14 (9.5%)	45 (38.5%)	59 (22.3%)		
Others	2 (1.4%)	21 (17.9%)	23 (8.7%)		
F75 to F75F transition (P1)	0 (0.0%)	53 (45.3%)	53 (20.1%)	**<0.001*****	<0.001
Family meals proportion (P1)				0.220	0.408
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 0.3	0 ; 1.0		
Food intolerance proportion (P1)				**0.009****	0.033
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 frequency below target (P1)				**0.022***	0.076
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 frequency above target (P1)					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F75 quantity below target (P1)				0.117	0.285
Median (Q1 ; Q3)	0 (0.0 ; 1)	0 (0.0 ; 1)	0 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 quantity above target (P1)				0.066	0.187
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		

Characteristic	Non-response to treatment in Phase				
	FALSE N = 147[1]	**TRUE** N = 117[1]	**Overall** N = 264[1]	**p-value**[2,3]	**q-value**[4]
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence below target (P1)				0.809	0.860
Median (Q1 ; Q3)	1 (0.0 ; 1)	1 (0.0 ; 1)	1 (0.0 ; 1)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence above target (P1)				**0.049***	0.152
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 0.8	0 ; 1.0		
Morbidity burden				0.089	0.234
Median (Q1 ; Q3)	1 (0.0 ; 2)	2 (1.0 ; 2)	1 (0.0 ; 2)		
Min ; Max	0 ; 4.0	0 ; 5.0	0 ; 5.0		
Shock	1 (0.8%)	3 (2.8%)	4 (1.7%)	0.335	0.518
Missing values	18	8	26		
Gastroenteritis	67 (51.9%)	66 (60.6%)	133 (55.9%)	0.182	0.388
Missing values	18	8	26		
Pneumonia	56 (43.4%)	51 (46.8%)	107 (45.0%)	0.602	0.705
Missing values	18	8	26		
Tuberculosis	1 (0.8%)	0 (0.0%)	1 (0.4%)	>0.999	>0.999
Missing values	18	8	26		
Digestive candidiasis	16 (12.4%)	19 (17.4%)	35 (14.7%)	0.275	0.446
Missing values	18	8	26		
HIV	0 (0.0%)	2 (1.8%)	2 (0.8%)	0.209	0.408
Missing values	18	8	26		
Severe malaria	45 (30.6%)	28 (23.9%)	73 (27.7%)	0.228	0.408
Neurological malaria	2 (1.6%)	3 (2.8%)	5 (2.1%)	0.663	0.751
Missing values	18	8	26		
Severe anemia	40 (31.0%)	30 (27.5%)	70 (29.4%)	0.557	0.701
Missing values	18	8	26		
Other anemia	40 (31.0%)	30 (27.5%)	70 (29.4%)	0.557	0.701
Missing values	18	8	26		
Death (P1)	0 (0.0%)	22 (56.4%)	22 (28.9%)	**<0.001*** **	<0.001
Missing values	110	78	188		
Death on day 1 (P1)	0 (0.0%)	7 (6.0%)	7 (2.7%)	**0.003****	0.013
Type of discharge (P1)				**<0.001*** **	<0.001

Characteristic	Non-response to treatment in Phase				
	FALSE N = 147[1]	**TRUE** N = 117[1]	**Overall** N = 264[1]	**p-value**[23]	**q-value**[4]
Recovered/Transferred	19 (51.4%)	13 (33.3%)	32 (42.1%)		
Absconded	18 (48.6%)	4 (10.3%)	22 (28.9%)		
Death	0 (0.0%)	22 (56.4%)	22 (28.9%)		
Missing values	110	78	188		

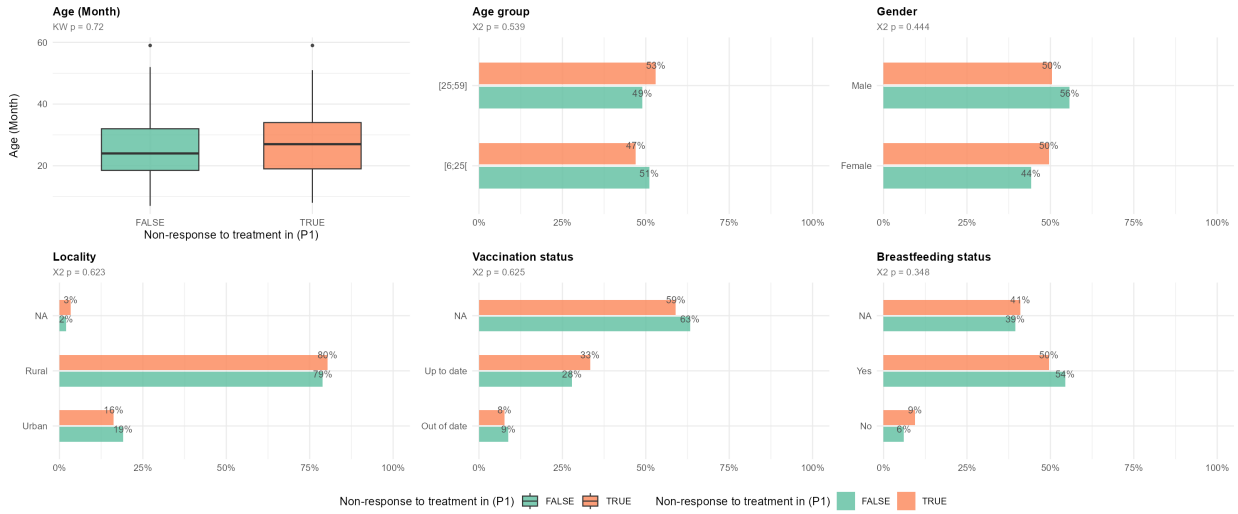
[1] n (%)

[2] Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test; NA; Fisher's Exact Test for Count Data with simulated p-value
(based on 2000 replicates)

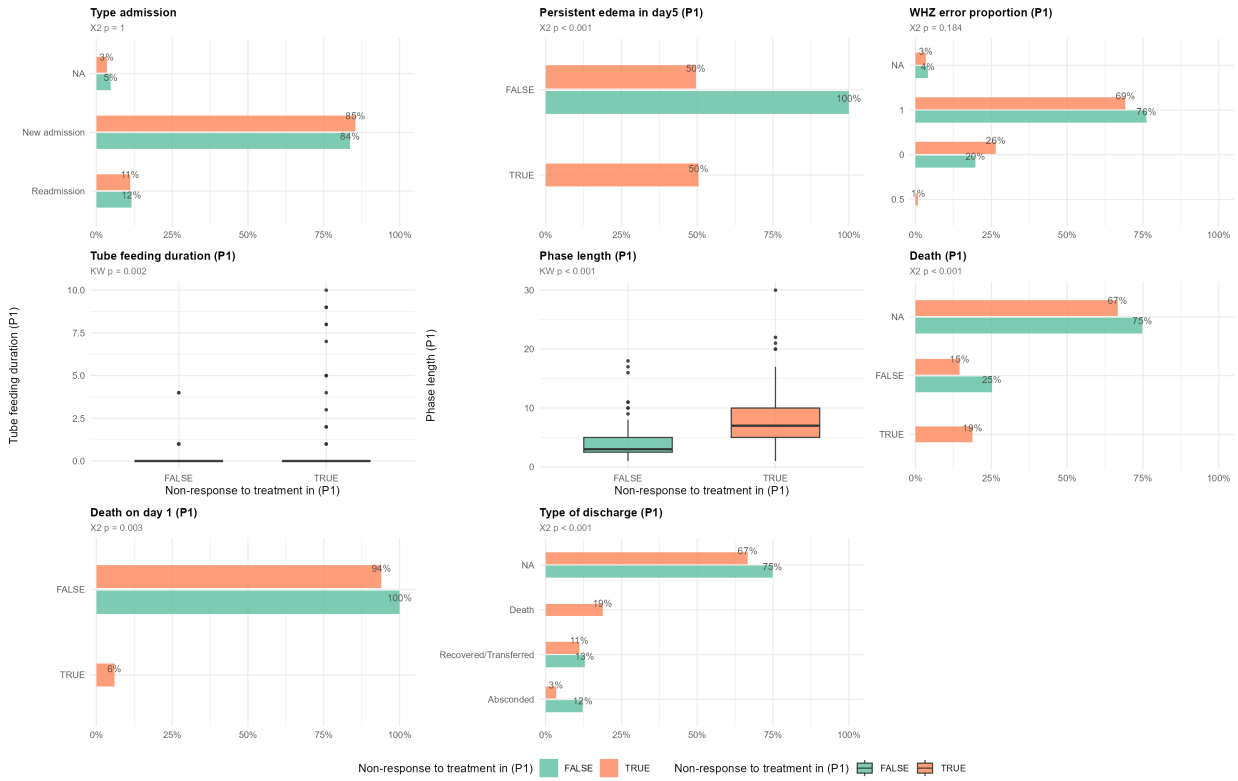
[3] *p<0.05; **p<0.01; ***p<0.001

[4] False discovery rate correction for multiple testing

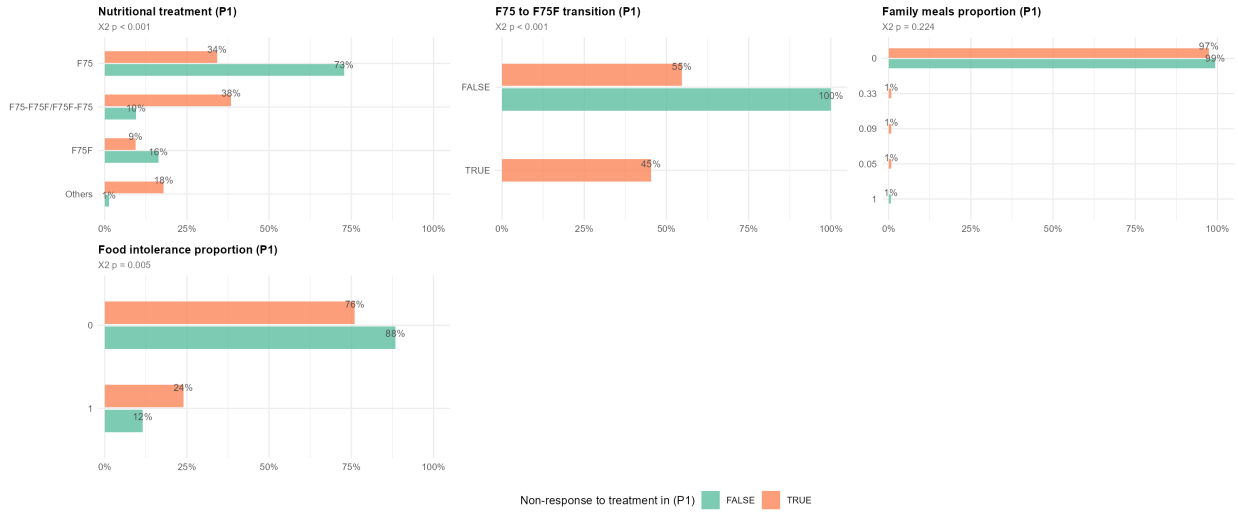
Sociodemographic



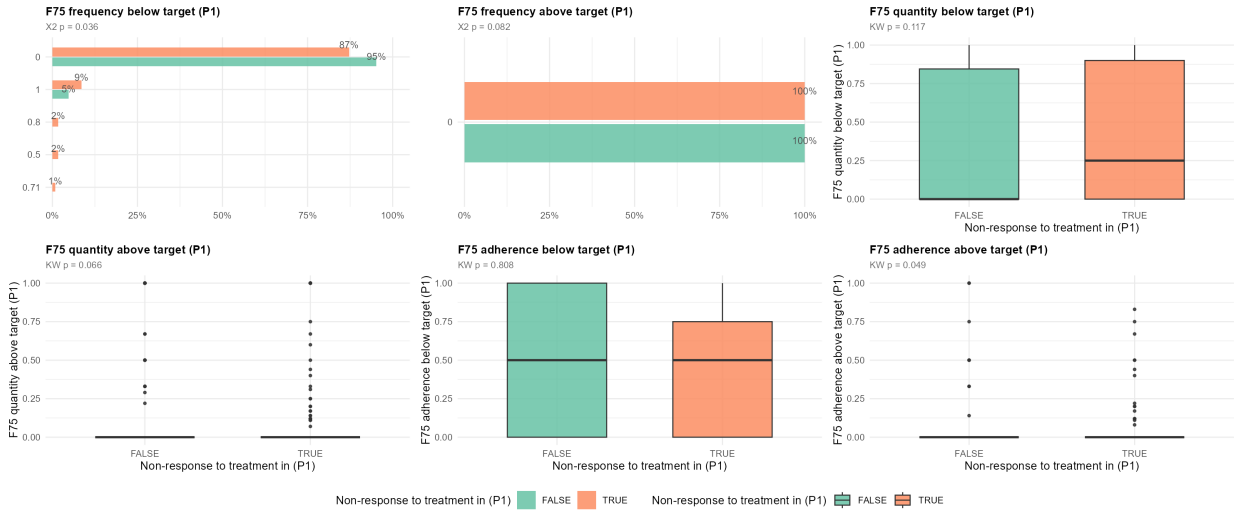
Clinical Admission and Severity



Nutritional Treatment

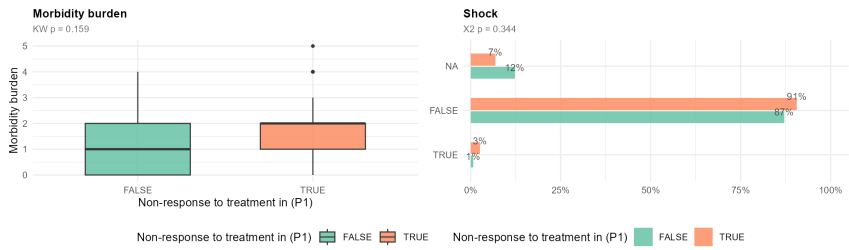


Feeding Compliance

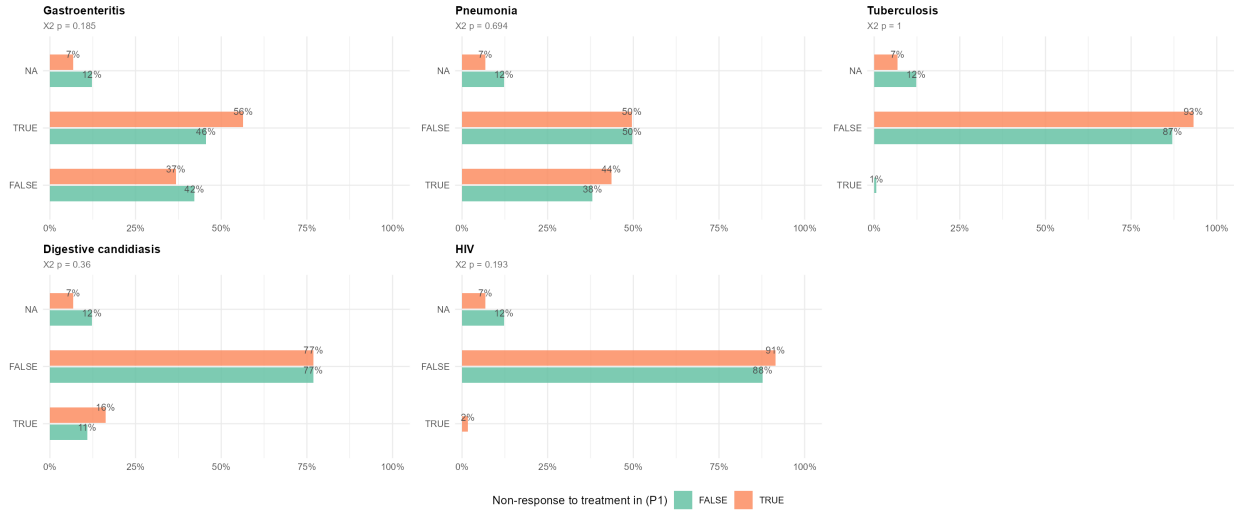


Morbidity and Comorbidities

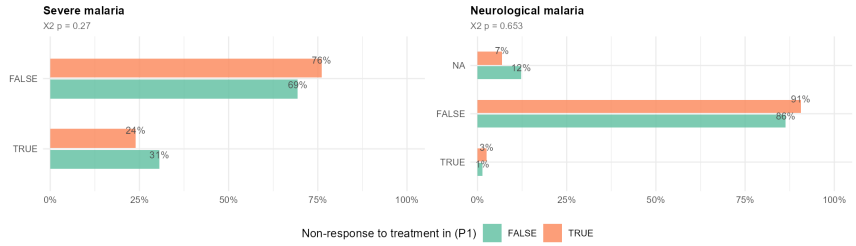
General



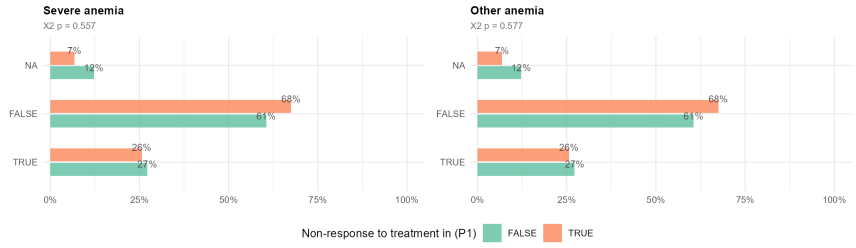
Infections and Parasitic



Malaria



Anaemia



Insert comments here.

Table 2: Results of the univariate logistic regression analysis between explanatory variables and outcome.

Predictor	Event N	OR	95% CI	p-value
F75 adherence below target (P1)				**0.041**
0	30	—	—	
]0;100]	87	1.73	1.02, 2.98	
Gastroenteritis				0.2
FALSE	43	—	—	
TRUE	66	1.42	0.85, 2.39	
Severe malaria				0.2
FALSE	89	—	—	
TRUE	28	0.71	0.41, 1.23	
Shock				0.2
FALSE	106	—	—	
TRUE	3	3.62	0.46, 73.8	
Digestive candidiasis				0.3
FALSE	90	—	—	
TRUE	19	1.49	0.73, 3.10	
Gender				0.4
Male	59	—	—	
Female	58	1.24	0.76, 2.02	
Age group				0.5
[25;59]	62	—	—	
[6;25[	55	0.85	0.52, 1.38	
Severe anemia				0.6
FALSE	79	—	—	
TRUE	30	0.84	0.48, 1.48	
Pneumonia				0.6
FALSE	58	—	—	
TRUE	51	1.15	0.69, 1.92	

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Insert comments here.

5. Multivariate logistic regression analyses

Table 3 : Results of the generalize logistic regression analysis at Phase 1.

Predictor	**N**	**Event N**	**OR**[1]	**95% CI**	**p-value**	**q-value**[2]	**VIF**
F75 adherence below target (P1)	238		*		0.021	**0.041**	1.0
0		24		—			
]0;100]		85	1.97	1.11, 3.56			
Severe malaria	238				0.13	0.13	1.0
FALSE		81		—			
TRUE		28	0.64	0.36, 1.13			

[1]*p<0.05; **p<0.01; ***p<0.001

[2]False discovery rate correction for multiple testing

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, VIF = Variance Inflation Factor

Insert comments here.